

Wilcoxon Signed-Rank Test

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1 Introduction

The Wilcoxon signed-rank test (paired version) is a non-parametric rank test for statistical hypothesis testing used to compare the locations of two populations using two matched samples. It does not assume that the data is normally distributed (which is why it is called non-parametric), and it is based on the differences between the two matched samples.

This test is frequently used in algorithm analysis, especially with stochastic algorithms. The goal is to test if the newest proposal behaves differently from previous works. For example, in a maximization problem, the goal is to test if the proposal tends to reach higher objective function values than the baseline algorithms.

2 Test Procedure

In order to execute the test, the following conditions must be met:

- **Paired data:** Data must be paired over the same set of instances or objects.
- **Pair independence:** Each pair of data must be completely independent from the rest.
- **Ordinal data:** Data needs to be sortable. It must be possible to determine which value is better or worse than another.
- **Symmetric difference:** The distribution of the differences between data pairs should ideally be symmetric. If this holds true, we can conclude whether the medians of the two populations differ. Otherwise, the test remains valid, but the conclusion shifts to stochastic dominance (i.e., one distribution systematically dominates the other). This is normally the case of interest to conclude if one algorithm performs better than another.

Also, as a statistical test, it relies on a null hypothesis (H_0) and an alternative hypothesis (H_1):

- H_0 : Assumes both algorithms' performance is the same, i.e., they belong to the same population.
- H_1 : Assumes both algorithms' performance differs, i.e., one performs better than the other.

The goal of the test is concluding that the performances of the algorithms differ. If the test does not succeed, the only valid conclusion is that there is not enough data to determine if both algorithms' performances differ. One must not reach the wrong conclusion by assuming that they behave the exact same way.

The test performs the following steps:

1. First, it computes the difference for each pair (X_i, Y_i) , obtaining:

$$V_i = X_i - Y_i$$

2. Then, it calculates the absolute value, $|V_i|$, while keeping a record of its original sign.
3. Additionally, it sorts the $|V_i|$ values in ascending order.
4. Next, it assigns a rank to each value, assigning 1 to the smallest $|V_i|$, 2 to the second smallest, and so on.
5. Finally, two values are calculated:

The sum of the ranks of the positive differences:

$$T^+ = \sum_{V_i > 0} \text{rank}(|V_i|)$$

And the sum of the ranks of the negative differences:

$$T^- = \sum_{V_i < 0} \text{rank}(|V_i|)$$

Therefore, $T^+ + T^-$ is equal to the sum of every assigned rank.

3 Handling Ties and Significance

During the assignment of ranks, it is common to encounter ties. If two algorithms produce the exact same objective function value for a given instance ($V_i = 0$), that instance is removed from the analysis, and the sample size N is reduced accordingly. If multiple differences share the same absolute value, they are assigned the average of the ranks they would have otherwise occupied. For example, if ranks 3 and 4 have the same value, they receive the rank 3.5.

Once T^+ and T^- are calculated, the test statistic T is defined as the minimum of both sums:

$$T = \min(T^+, T^-)$$

To determine the statistical significance of T , the procedure depends on the sample size N (excluding the zero differences). For large samples ($N > 20$), the distribution of the T statistic approximates a normal distribution. The expected mean μ_T and standard deviation σ_T under the null hypothesis are calculated as follows:

$$\mu_T = \frac{N(N+1)}{4}$$

$$\sigma_T = \sqrt{\frac{N(N+1)(2N+1)}{24}}$$

Using these values, a Z -score is computed to find the final p -value:

$$Z = \frac{T - \mu_T}{\sigma_T}$$

Finally, the computed p -value is compared against the predefined significance level (α). If $p \leq \alpha$, the null hypothesis is rejected, demonstrating that the algorithms do not perform equally.

Notice that the p -value represents the probability of obtaining the observed results assuming the null hypothesis is true. If its value is small enough (determined by α), a highly improbable event has occurred; therefore, the null hypothesis can be safely discarded.

Once the p -value is calculated, and in the case where $p < \alpha$, the direction of the performance difference needs to be analyzed. For this purpose, the confidence interval is examined. If the interval does not contain zero (i.e., it lies entirely on one side), there is statistical evidence that the proposed algorithm is expected to systematically reach better objective function values than the previous algorithm.

4 Example

To illustrate the application of the Wilcoxon signed-rank test, along with the handling of ties and zero differences, consider a hypothetical scenario comparing the proposed GRASP algorithm against the baseline IG algorithm. Assuming a maximization problem, both algorithms are executed over six instances, yielding the objective function values presented in Table 1.

Table 1: Objective function values for GRASP and IG across six instances.

Instance	GRASP (X_i)	IG (Y_i)	Difference (V_i)	Absolute ($ V_i $)
Instance 1	94	90	+4	4
Instance 2	85	85	0	0
Instance 3	78	80	-2	2
Instance 4	92	90	+2	2
Instance 5	88	83	+5	5
Instance 6	99	91	+8	8

Following the established test procedure:

1. **Calculate differences:** The differences $V_i = X_i - Y_i$ and their absolute values are computed.
2. **Handle zero differences:** Instance 2 results in a tie ($V_2 = 0$). Consequently, it is removed from the analysis, reducing the sample size from $N = 6$ to $N = 5$.
3. **Sort and rank:** The remaining absolute differences are sorted in ascending order: 2, 2, 4, 5, 8.
4. **Handle absolute value ties:** The value 2 appears twice, occupying the theoretical ranks 1 and 2. They are assigned the average rank of 1.5. The remaining values receive ranks 3, 4, and 5, respectively.
5. **Calculate rank sums (T^+ and T^-):** The sum of the ranks for the positive differences (+2, +4, +5, +8) is:

$$T^+ = 1.5 + 3 + 4 + 5 = 13.5$$

The sum of the ranks for the negative differences (-2) is:

$$T^- = 1.5$$

The test statistic T is defined as the minimum of the two sums:

$$T = \min(13.5, 1.5) = 1.5$$

To determine the statistical significance, the expected mean and standard deviation are calculated for $N = 5$:

$$\mu_T = \frac{5(5+1)}{4} = 7.5$$

$$\sigma_T = \sqrt{\frac{5(5+1)(2(5)+1)}{24}} = \sqrt{13.75} \approx 3.708$$

The standardized Z -score is then computed:

$$Z = \frac{1.5 - 7.5}{3.708} \approx -1.618$$

Evaluating $Z = -1.618$ in a standard normal distribution table yields a p -value of approximately 0.105. If the significance level is set to $\alpha = 0.05$, then $p > \alpha$. In this small-scale illustrative example, the null hypothesis cannot be rejected. However, in comprehensive experiments (e.g., $N = 200$), a continuous accumulation of positive ranks would lead to a significant p -value ($p < \alpha$), thus demonstrating the superiority of the proposed algorithm.